#8 (2) $\theta = \frac{\pi}{6}$ 에서의 접선 식 ? ← $x = a\left(1+\frac{\sqrt{3}}{2}\right)\cdot\frac{\sqrt{3}}{2}$, $y = a\left(1+\frac{\sqrt{3}}{2}\right)\cdot\frac{1}{2}$

$r = a(1+\cos\theta)$ // $\left(a\left(1+\frac{\sqrt{3}}{2}\right), \frac{\pi}{6}\right)$

$r = a(1+\cos\theta) = f(\theta)$

$x = a\cdot\cos\theta(1+\cos\theta) = a\cos\theta + a\cos^2\theta \rightarrow \frac{1}{2}a\sin 2\theta$

$y = a\sin\theta(1+\cos\theta) = a\sin\theta + a\sin\theta\cos\theta$

$\frac{dy}{dx} = \frac{a\cos\theta + a\cos 2\theta}{-a\sin\theta - 2a\cos\theta\sin\theta} = \frac{\cos\theta + \cos 2\theta}{-\sin\theta - \sin 2\theta}$

$\theta = \frac{\pi}{6}$ 에서 $\frac{dy}{dx} = \frac{\frac{\sqrt{3}}{2}+\frac{1}{2}}{-\frac{1}{2}-\frac{\sqrt{3}}{2}} = \frac{\sqrt{3}+1}{-1-\sqrt{3}} = -1$

$\therefore\ y - \frac{a}{2}\left(1+\frac{\sqrt{3}}{2}\right) = -\left(x - \frac{\sqrt{3}}{2}a\left(1+\frac{\sqrt{3}}{2}\right)\right)$ //

#9 (1)

대명 ① $r = 4\cos 3\theta$, $r = 4\sin 3\theta$ ② $P\left(2\sqrt{2}, \frac{\pi}{12}\right)$

① $4\cos 3\cdot\frac{\pi}{12} = 2\sqrt{2}$

② $4\sin 3\cdot\frac{\pi}{12} = 2\sqrt{2}$

∴ 만남

(2) $\tan\alpha$ //

$\tan\psi = \frac{r}{r'} = \frac{4\cos 3\theta}{-12\sin 3\theta} = -\frac{1}{3}\cot 3\theta$

$\tan\psi' = \frac{r}{r'} = \frac{4\sin 3\theta}{12\cos 3\theta} = \frac{1}{3}\tan 3\theta$

$\theta = \frac{\pi}{12}$ 에서

$\tan\psi = -\frac{1}{3}\cot\frac{\pi}{4} = -\frac{1}{3}$

$\tan\psi' = \frac{1}{3}$ //

$\therefore\ \tan\alpha = \left|\frac{-\frac{2}{3}}{1-\frac{1}{9}}\right| = \frac{\cancel{2}}{\cancel{3}}\cdot\frac{\cancel{3}^{3}}{\cancel{8}_{4}}$

$= \frac{3}{4}$ ∎

#11 $r = 4\cos 3\theta$

